SIMATS ENGINEERING
ASSESSMENT TOOL 2:

Case Study

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)

Dave's Taxonomy: Precision (Level 3)
Question Count: 3

Total Marks: 30

Code of Conduct

I Rakshana · A.R (192126002 7) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A.R

Assessment Tasks

Case Study

1. **Case Study 1: Smart Retail Inventory Management System**

   A retail chain implements a cloud-based inventory system connected to store devices.Clients (POS systems and dashboards) communicate via APIs to centralized cloud services. Cloud storage stores product data, transaction logs, and analytics datasets. The system uses platform services for analytics and demand forecasting. Secure access is maintained through VPNs, firewalls, and API gateways. This solution integrates infrastructure, services, and web applications for efficient operations.

2. **Case Study 2: Cloud-Based Document Collaboration System**

   An enterprise deploys a document collaboration tool similar to Google Docs. Users access the platform via browsers, enabling real-time editing and sharing. Documents are stored in distributed cloud storage with version control mechanisms. Web APIs enable synchronization across multiple clients and devices.

   Network infrastructure ensures low latency and high availability globally. Security includes encryption, identity management, and access permissions for collaboration.

3. **Case Study 3: Online Food Ordering Platform**

A startup builds a cloud-based food ordering system accessible via web browsers and mobile clients. Frontend clients interact with backend services through RESTful Web APIs hosted on a cloud platform.

Cloud storage is used to store menu images, invoices, and customer data securely. The infrastructure uses virtual machines and managed Kubernetes for scalability and reliability. Security is enforced through authentication tokens, HTTPS, and role-based access control. The system demonstrates integration of clients, APIs, storage, and platform services in real time.

Case Study Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Understanding of Case	25%	Clearly identifies all technical and business requirements	Identifies most requirements	Basic understanding only	Poor understanding of case
Application of Concepts	25%	Applies suitable cloud concepts and tools effectively	Good application with minor issues	Partial application	Weak or incorrect application
Analysis	20%	Strong analysis with clear trade-off reasoning	Reasonable analysis	Limited analysis	Minimal analysis
Solution Design	20%	Solution is feasible, practical, and well justified	Reasonable solution with minor gaps	Basic solution	Unclear solution
Clarity	10%	Well-structured and easy to follow	Mostly clear	Somewhat unclear	Poorly organized

1. Case Study 1: Smart Retail Inventory Management System:

1) Introduction:

A Smart Retail Management system in a cloud-based solution that helps retail stores manage product details, sales transactions, stock availability, & demand forecasting. It connects Point of Sale (POS) systems, store devices, & dashboards to centralized cloud services.

2) Cloud Architecture & Components:

The system consists of the following components:

1) Client layer:

POS systems and dashboards allow employees to access inventory information.

2) Web APIs:

Connect Store devices with centralized cloud services.

3) Cloud Storage:

Stores product data, transaction logs, and analytics data.

4) Platform Services:

Analyse sales data & forecast product demand.

5) Security Layer:

VPNs, firewalls, & API gateways Protect the system.

Working:

1) Customers Purchase products through POS systems.
2) Transaction details are sent to cloud services through APIs.
3) The inventory database is updated automatically.
4) Cload storage maintains Product & transaction information.
5) Analytics services predict future demand using historical scales data.
6) Managers use dashboards to monitor stock & make decisions.

Security Measures:

- VPN provides secure communication between stores & cloud servi
- Firewalls Prevent unathorized network.
- API gateways control API requests.
- Authentication & authorization restrict access to authorized
- Encryption protects sensitive data.

Advantages:

- Real-time inventory tracking.
- Reduced stock shortages & overstocking.
- Improved demand forecasting.
- Centralized data management
- Better business decisions.

2. Case 2: Cloud - Based Document Collaboration System.

Introduction:

A cloud - Based Document System is an online application similar to Google Docs. It allows multiple users to create, edit, & share documents in real time using web browsers and cloud services.

Architecture & Components:

1) Client Layer: Web browsers & devices provide access to documents.
2) Application Layer: Handles document editing & sharing.
3) Web APIs: Synchronize document changes across multiple clients.
4) Distributed Cloud storage: Stores documents & version history.
5) Network Infrastructure: Provides low latency & high availability.
6) Security Layer: Uses encryption, identity management, & access permissions.

Working:

1) Users log in through a web browser.
2) Documents are retrieved from cloud storage.
3) Users edit documents through the application.
4) Web APIs transmit updates to the cloud server.
5) Changes are synchronized with other authorized users.
6) Version control maintains previous document versions for recovery.

Security Measures:

- Encryption protects documents during transmission & storage.
- Identity management verifies users.
- Access permissions control who can view or edit documents.
- Authentication prevents unauthorized access.
- Version control helps recover earlier document versions.

Advantages:

- Real-time collaboration.
- Access from anywhere.
- Easy document sharing
- Automatic synchronization.
- Document history & recovery
- Improved teamwork & productivity.

3. Case 3: Online Food Ordering Platform.

Introduction:

An Online Food Ordering platform is a cloud-based application that allow customers to browse menus & order food through websites or mobile applications. It uses RESTful Web APIs, cloud storage, virtual machines, & managed Kubernetes to provide scalability & reliability.

Architecture & Components:

1) Frontend layer:

Web browsers & mobile applications provide the user interface.

2) RESTful Web APIs:

Connect frontend clients with backend services.

3) Bakend Services:

Process menu requests & customer data.

4) Cloud Storage:

Stores menu images, invoices, & customer data.

5) Virtual Machines:

Provide computing resources for the application.

6) Managed Kubernetes:

Manages containerized services & supports scalability.

7) Security Layer:

Uses authentication tokens, HTTPs, & role-based access control.

Working:

1) Customers open the website or mobile application.
2) The frontend requests menu details through RESTful APIs.
3) Backend services retrieve the required menu information.
4) Customers select food items & place orders.
5) Backend services process orders & store relevant information.
6) Cloud storage maintains invoices, menu images & customer data.
7) Virtual machines & Kubernetes run the application services
8) The cloud infrastructure scales to handle increased customer requests.

Security Measures:

- Authentication tokens verify user identity.
- HTTPS encrypts communication between clients & servers.
- Role-Based Access Control (RBAC) restricts access according to user roles.
- API security. Prevents unauthorized requests.